GM - 2024

Q7544109
Question Paper SL No.

2024
MATHEMATICS

3 Hours 15 Minutes

(First 15 minutes for reading the question paper)

Full Marks — { 90 — **For Regular Candidates**
100 — **For External Candidates**

Special credit will be given for answers which are brief and to the point.
Marks will be deducted for spelling mistakes, untidiness & bad handwriting.

[The answers of the Question Nos. 1, 2, 3, 4 are to be written at the beginning of the answer-script mentioning the question numbers in the serial order. Necessary calculation and drawing must be given on the right hand side by drawing margins on the first few pages on the answer-script. Tables and Calculators of any type are not allowed. Approximate value of π may be taken as $\frac{22}{7}$, if necessary. Graph paper will be supplied with question paper. Arithmetic problems may be solved by algebraic method.]

[Alternative Question No. 11 is given for Sightless Candidates on Page No. 16]

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1. Choose the correct option in each case from the following: $1 \times 6 = 6$

(i) If the rate of simple interest and compound interest are both 10% per annum, then the ratio of simple interest and compound interest in second year of a certain principal will be—

(a) 20 : 211 (b) 10 : 111

(c) 5 : 61 (d) 1 : 11

(ii) If one root of the equation $ax^2 + abcx + bc = 0$ $(a \neq 0)$ is reciprocal to the other, then—

(a) $abc = 1$ (b) $b = ac$

(c) $bc = 1$ (d) $a = bc$

(iii) Two circles of diameter 5 cm and 7 cm touch each other internally, the distance between their centres will be—

(a) 1 cm (b) 2 cm

(c) 3 cm (d) 4 cm

(iv) The minimum value of $\tan\theta + \cot\theta$ is—

(a) 0 (b) 2

(c) -2 (d) 1

(v) If the height and base area of a solid hemisphere and a solid cylinder are same, then the ratio of their volume is—

(a) 1 : 3 (b) 1 : 2

(c) 2 : 3 (d) 3 : 4

(vi) If A and M be the arithmetic mean and median of first ten natural numbers, then relation between them will be—

(a) $A > M$ (b) $A < M$

(c) $A = \frac{1}{M}$ (d) $A = M$

2. Fill up the blanks (any **five**): $1 \times 5 = 5$

(i) The equation $(P-3)x^2 + 5x + 10 = 0$ will not be a quadratic equation for $P =$ .....

.

(ii) The sum of Principal and Compound interest in certain time is called .....

(iii) The corresponding sides of two similar triangles are .....

(iv) If $\sin(\theta - 30^\circ) = \frac{1}{2}$, then $\cos\theta =$ .....

(v) If V be the volume, R be the radius of the base and H be the height of a right circular cone, then $H =$ .....

(vi) The numbers $8, 9, 12, 17, x+2, x+4, 30, 34, 39$ are in increasing order and if the median of these numbers is 24, then $x =$ .....

3. Write True or False (any **five**): $1 \times 5 = 5$

(i) In a partnership business, the ratio of the capitals of three members is $a : b : c$ and the ratio of time of investment of their capitals is $x : y : z$. The rate of their profit will be $ax : by : cz$.

(ii) If $a \propto b$, $b \propto \frac{1}{c}$ and $c \propto d$ then $a \propto \frac{1}{d}$.

(iii) If two chords of a circle are at an equal distance from the centre of that circle, then those two chords will be always parallel.

(iv) The hour hand of a clock rotates through an angle $\frac{\pi}{6}$ radian in 2 hours.

(v) The ratio of whole surfaces of a solid sphere and a solid hemisphere of equal radius is $2 : 1$.

(vi) The average of n numbers is $\overline{x}$. If the sum of first $(n-1)$ numbers is K, then n^{th} number is $(n-1)\overline{x} + K$.

4. Answer the following questions (any **ten**): $2 \times 10 = 20$

(i) In how many years will Rs. 500 get Rs. 105 compound interest at 10% per annum compound interest?

(ii) In a partnership business, ratio of capitals of Ila, Rahima and Bela is $3 : 8 : 5$. If profit of Ila is Rs. 600 less than profit of Bela. Find the total profit in the business.

(iii) If the roots of the equation : $x^2 - 22x + 105 = 0$ are α and β, then find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$.

(iv) If $(3x - 2y) : (3x + 2y) = 4 : 5$ then what is the value of $(x + y) : (x - y)$?

(v) BOC is a diameter of a circle whose centre is O. ABCD is a cyclic quadrilateral. If $\angle ADC = 110^\circ$ find $\angle ACB$.

(vi) In trapezium ABCD, BC || AD and AD = 4 cm, two diagonals AC & BD meet at O in such a way that $\frac{AO}{OC} = \frac{DO}{OB} = \frac{1}{2}$, then find the length of BC.

(vii) In ΔABC, $\angle ABC = 90^\circ$, AB = 6 cm, BC = 8 cm, then find the circumradius of the triangle ABC.

(viii) If $r\cos\theta = 2\sqrt{3}$ and $r\sin\theta = 2$, then find the value of r and θ when $0^\circ < \theta < 90^\circ$.

(ix) If $\sin(A + B) = 1$ and $\cos(A - B) = 1$, then find the value of $\cot 2A$, $0^\circ \leq (A + B) \leq 90^\circ$ and $A \geq B$.

(x) How much percent of the curved surface area of a sphere will be increased if the radius is doubled.

(xi) The length of the diagonal of each face of a cube is $6\sqrt{2}$ cm. Find the total surface area of the cube.

(xii) The mean of a frequency distribution is 7 and $\sum f_i x_i = 140$, then find the value of $\sum f_i$.

5. Answer any **one** question : 5

(i) After retirement from service, Gobinda Babu got Rs. 5,00,000. He deposited a part of it in post office at 7.2% simple interest per annum and he deposited other part in a bank at 6% simple interest per annum. Every year he got Rs. 33,600 in total as interest from bank and post office. Find the amounts he deposited in bank and post office separately.

(ii) Aman has taken a loan of Rs. 25,000 for 3 years, in such a way what the compound interest for 1st year, 2nd year and 3rd year is 4%, 5% and 6% p.a. respectively. Find the amount be deposited after 3 years.

6. Answer any **one** question : 3

(i) Speed of A is 1m/sec more than that of B. A reaches 2 seconds earlier than B in 180m sprint competition. Find the speed of B in m/sec.

(ii) Solve: $(2x+1) + \frac{3}{(2x+1)} = 4, \left(x \neq -\frac{1}{2}\right)$

7. Answer any **one** question: 3

(i) If $(\sqrt{a} + \sqrt{b}) \propto (\sqrt{a} - \sqrt{b})$ then show that $(a+b) \propto \sqrt{ab}$

(ii) If $x = \sqrt{3} + \sqrt{2}$ and $y = \frac{1}{x}$ then find $\left(x + \frac{1}{x}\right)^2 + \left(\frac{1}{y} - y\right)^2$.

8. Answer any **one** question: 3

(i) If $\frac{x}{y+z} = \frac{y}{z+x} = \frac{z}{x+y}$, then show that each ratio is $\frac{1}{2}$ or -1.

(ii) If a, b, c are in continued proportion, then prove that $\frac{1}{b} = \frac{1}{b-a} + \frac{1}{b-c}$.

9. Answer any **one** question: 5

(i) Prove that in any circle, angles in the same segment are equal.

(ii) If two tangents are drawn to a circle from a point outside it, then prove that the line segments joining the point of contact and the exterior point are equal and they subtend equal angles at the centre.

10. Answer any **one** question: 3

(i) $ABCD$ is a circumscribed quadrilateral of a circle with centre O. Show that

$AB + CD = AD + BC$

(ii) In a triangle PQR, $\angle P = 90^\circ$ and PS is perpendicular to QR.

Then prove that $\frac{1}{PS^2} - \frac{1}{PQ^2} = \frac{1}{PR^2}$

11. Answer any **one** question: 5

(i) Draw a circle of radius 4 cm. Draw a tangent on this circle from an external point at a distance 9 cm from the centre of the circle.

(ii) Draw a right angled triangle whose two adjacent sides of the right angle are 4 cm and 5 cm. Construct the circumcircle of this triangle.

12. Answer any **two** questions: $3 \times 2 = 6$

(i) The difference of acute angles of a right angled triangle is 72°. Determine acute right angles in circular measure.

(ii) If $5\sin^2\theta + 4\cos^2\theta = \frac{9}{2}$, then from this relation find the value of $\tan\theta$

(iii) If $\sin 17^\circ = \frac{x}{y}$ then show that $\sec 17^\circ - \sin 73^\circ = \frac{x^2}{y\sqrt{y^2-x^2}}$.

13. Answer any **one** question: 5

(i) The angle of elevation of the top of a pillar of height h from two points in the same horizontal line through the foot of the pillar and in the same side of the pillar are θ and ϕ respectively. Find

the distance between two points.

(ii) Two pillars of equal heights are at the point A and B on the opposite side of the road which is 120m wide. From a point C on the line joining the foots of the pillars, the angle of elevation of the top of the pillar at A and B are 60° and 30° respectively. Find AC.

14. Answer any **two** questions: $4 \times 2 = 8$

(i) The lower part of an ice-cream is conical and upper part is hemispheric with same circular base. If the height of the cone is 9 cm and radius of the base 2.5 cm. Find the volume of the ice-cream.

(ii) Difference of outer and inner curved surface area of a hollow cylindrical pipe is 44 sq. cm and length is 14 cm. If the Volume of the material of the pipe be 99 cubic cm. Find inner and outer radius.

(iii) If 75 buckets of water of equal measure are taken out from the water of a cubical water-filled-tank, then $\frac{2}{5}$th of the water remains in the tank. If the length of one edge of the tank is 1.5 metres, then calculate the quantity of water in litre that can be held in each bucket.

15. Answer any **two** questions : $2 \times 4 = 8$

(i) Find the mode from the following data.

Class Interval	0–5	5–10	10–15	15–20	20–25	25–30	30–35	35–40
Frequency	2	6	10	16	22	11	8	5

(ii) Find the mean by any suitable method from the following frequency distribution table.

Class Interval	85–105	105–125	125–145	145–165	165–185	185–205
Frequency	3	12	18	10	5	2

(iii) Find the median of the data from the following distribution.

Marks Obtained	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60

No. of Students	8	15	29	42	60	70

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